

The Exact Correction Factor Between the de Sitter Acceleration Scale and the MOND Acceleration Constant: An Empirical Bridge to Milgrom's Thermal Framework

Dat Nguyen Independent Researcher Dtnghuyen09@gmail.com ORCID: 0009-0001-2514-7768

Novelty, Significance, and Value

What we knew before

In 1999, Milgrom showed that an accelerated observer in de Sitter space sees Unruh-like radiation whose temperature difference $\Delta T = T(a) - T(0)$ has exactly the MOND functional form: linear in acceleration for $a \gg a_{dS}$ (Newtonian) and quadratic for $a \ll a_{dS}$ (deep-MOND). This was a striking connection between MOND and horizon thermodynamics. But the transition scale was wrong: $a_{dS} = cH_0\sqrt{\Omega_\Lambda} = 5.42 \times 10^{-10} \text{ m/s}^2$ is 4.6 times larger than the observed $a_0 = 1.20 \times 10^{-10} \text{ m/s}^2$. Milgrom called the agreement "of order unity" and left the factor unidentified. For 25 years, no one has provided the exact correction.

What this paper shows

The empirical formula $a_0 = cH_0(\Omega_\Lambda/\Omega_m)^{1/6}/(2\pi)$ (Nguyen 2026a) gives the exact ratio:

$$a_{dS} / a_0 = 2\pi \times \Omega_\Lambda^{1/3} \times \Omega_m^{1/6} = 4.57$$

Each component has a physical interpretation: 2π converts from horizon radius to circumference, $\Omega_\Lambda^{1/3}$ encodes vacuum energy in $D = 3$ dimensions, and $\Omega_m^{1/6}$ encodes matter dilution through a geometric mean. This is an algebraic result — exact given the empirical a_0 formula — not a derivation from first principles.

What this paper does not show

This paper does **not** derive the thermal temperature formula for an accelerated observer in Λ CDM spacetime. The Deser-Levin (1997) formula $T(a) = (\hbar/(2\pi k_B c))\sqrt{(a^2 + a_{dS}^2)}$ was derived for pure de Sitter space. Replacing a_{dS} with a_0 is empirically motivated, not theoretically derived. Whether the Λ CDM temperature formula takes this form with a_0 replacing a_{dS} is an open problem in quantum field theory in curved spacetime.

Additionally, the thermal interpolation function implied by the Milgrom (1999) framework has a known factor-of-2 discrepancy with standard MOND in the deep-MOND limit (Section 5). This discrepancy is present in Milgrom's original work and is not resolved here.

Implications

Despite these caveats, the exact correction factor is a concrete result that constrains any future thermal derivation of MOND in Λ CDM: whatever that derivation produces, it must recover $a_0 = a_{\text{dS}} / (2\pi \times \Omega_{\Lambda}^{1/3} \times \Omega_{\text{m}}^{1/6})$. The decomposition into circumference, dimensionality, and matter factors provides specific targets for theoretical work.

1. Introduction

Milgrom (1983) introduced MOND with the acceleration constant $a_0 \approx 1.2 \times 10^{-10} \text{ m/s}^2$ as a free parameter and immediately noted its approximate coincidence with cH_0 . In 1999, Milgrom proposed a physical origin through the Unruh effect in de Sitter space: an observer with proper acceleration a sees thermal radiation at temperature (Deser & Levin 1997):

$$T(a) = (\hbar/(2\pi k_{\text{B}} c)) \sqrt{a^2 + a_{\text{dS}}^2} \quad \dots (1)$$

where $a_{\text{dS}} = c^2/(\Lambda/3)^{1/2}$. The temperature difference $\Delta T(a) = T(a) - T(0)$ has the MOND functional form, with the transition at $a \sim a_{\text{dS}}$. This connected MOND to horizon thermodynamics conceptually, but the transition scale $a_{\text{dS}} = cH_0\sqrt{\Omega_{\Lambda}} = 5.42 \times 10^{-10} \text{ m/s}^2$ is 4.6 times larger than the observed a_0 .

Other authors have explored related connections. Klinkhamer (2011) derived MOND from a minimum temperature in entropic gravity with $a_0 \propto cH_0$ but no exact factor. Verlinde (2016) obtained MOND-like behavior from de Sitter entropy with $a_0 \sim cH_0$. McCulloch (2007) proposed modified inertia from a Hubble-scale Casimir effect on Unruh radiation, obtaining $a_0 \sim cH_0$ using the Hubble diameter as a wavelength cutoff. Volovik (2024) argued that the local de Sitter temperature is $T = H/\pi$, twice the Gibbons-Hawking temperature $T_{\text{GH}} = H/(2\pi)$, raising unresolved questions about the correct factor even in pure de Sitter space. In all cases, the exact numerical relationship between a_{dS} and a_0 has remained unspecified.

In this paper, we show that the empirical formula $a_0 = cH_0(\Omega_{\Lambda}/\Omega_{\text{m}})^{1/6}/(2\pi)$ (Nguyen 2026a), tested against 3,389 SPARC rotation curve measurements, provides the exact correction factor. We decompose it, interpret its components, and discuss both its strengths and its limitations as a bridge between the empirical a_0 and Milgrom's thermal framework.

2. The Pure de Sitter Framework

2.1 Temperature in de Sitter space

An inertial observer in de Sitter space sees the Gibbons-Hawking temperature:

$$T_{\text{GH}} = \hbar H_{\text{dS}} / (2\pi k_{\text{B}} c) \quad \dots (2)$$

An observer with constant proper acceleration a sees (Deser & Levin 1997):

$$T(a) = (\hbar / (2\pi k_{\text{B}} c)) \sqrt{a^2 + a_{\text{dS}}^2} \quad \dots (3)$$

where $a_{\text{dS}} = c H_{\text{dS}} = c^2 (\Lambda/3)^{1/2}$. This combines the Unruh effect with the Gibbons-Hawking effect in quadrature.

Important caveat: Equation (3) is rigorously derived only for pure de Sitter space with the static metric $ds^2 = -(1 - H^2 r^2) c^2 dt^2 + (1 - H^2 r^2)^{-1} dr^2 + r^2 d\Omega^2$. The parameter a_{dS} is the surface gravity of the de Sitter horizon, determined by the spacetime geometry. It is not a free parameter — it is fixed by the metric.

2.2 The MOND functional form

The temperature difference:

$$\Delta T(a) = T(a) - T(0) = (\hbar / (2\pi k_{\text{B}} c)) [\sqrt{a^2 + a_{\text{dS}}^2} - a_{\text{dS}}] \quad \dots (4)$$

Two limits:

- $a \gg a_{\text{dS}}$: $\Delta T \approx (\hbar / (2\pi k_{\text{B}} c)) \times a$ — linear in a (Newtonian inertia).
- $a \ll a_{\text{dS}}$: $\Delta T \approx (\hbar / (4\pi k_{\text{B}} c)) \times a^2 / a_{\text{dS}}$ — quadratic in a (MOND-like modified inertia).

If inertia is proportional to ΔT (Milgrom 1999), the dynamics transition from Newtonian to MONDian at $a \sim a_{\text{dS}}$. The functional form is correct.

2.3 The quantitative problem

In our universe ($\Omega_{\Lambda} = 0.685$, $H_0 = 67.4$ km/s/Mpc):

$$a_{\text{dS}} = c H_0 \sqrt{\Omega_{\Lambda}} = 5.42 \times 10^{-10} \text{ m/s}^2$$

$$\text{Observed: } a_0 = 1.20 \times 10^{-10} \text{ m/s}^2$$

Discrepancy: $a_{dS}/a_0 = 4.57$

This factor has been acknowledged but unexplained for 25 years.

3. The Exact Correction Factor

3.1 The empirical formula

The formula $a_0 = cH_0(\Omega_\Lambda/\Omega_m)^{1/6}/(2\pi)$ (Nguyen 2026a) produces $a_0 = 1.186 \times 10^{-10} \text{ m/s}^2$, tested against 3,389 SPARC data points with $\Delta\chi^2 = 4,895$ relative to $cH_0/(2\pi)$.

3.2 The ratio

$$a_{dS}/a_0 = cH_0\sqrt{\Omega_\Lambda} / [cH_0(\Omega_\Lambda/\Omega_m)^{1/6}/(2\pi)]$$

$$= 2\pi \times \Omega_\Lambda^{1/2} / (\Omega_\Lambda/\Omega_m)^{1/6}$$

$$= 2\pi \times \Omega_\Lambda^{1/2 - 1/6} \times \Omega_m^{1/6}$$

$$= 2\pi \times \Omega_\Lambda^{1/3} \times \Omega_m^{1/6} \quad \dots (5)$$

With $\Omega_\Lambda = 0.685$ and $\Omega_m = 0.315$:

$$a_{dS}/a_0 = 2\pi \times 0.8815 \times 0.8249 = 4.569 \quad \dots (6)$$

This is an algebraically exact result given the empirical a_0 formula. It does not depend on any assumption about the thermal mechanism.

3.3 Decomposition of the factor

The three components of Equation (5):

Factor 1: $2\pi = 6.283$. This is the largest component. In Danny's framework (Nguyen 2026c), the 2π represents the circumference-to-radius ratio of the Hubble sphere. The de Sitter formula uses the horizon radius as its length scale ($a_{dS} = c/\ell_\Lambda$ where ℓ_Λ is the de Sitter radius). The empirical a_0 uses the circumference ($a_0 \sim c^2/(2\pi\ell_\Lambda)$ with corrections). Whether the correct physical scale for the thermal effect is the circumference rather than the radius is a question for the Λ CDM thermal calculation that has not yet been performed.

Factor 2: $\Omega_\Lambda^{1/3} = 0.882$. This reflects the vacuum energy density in a universe where $\Omega_\Lambda < 1$. In pure de Sitter space, $\Omega_\Lambda = 1$ and this factor vanishes. The $1/3$ power corresponds to the

number of spatial dimensions $D = 3$. In the algebraic decomposition (Nguyen 2026b), this arises from matter diluting as $(1+z)^D$.

Factor 3: $\Omega_m^{1/6} = 0.825$. This is the matter correction. The $1/6 = 1/(2D)$ power combines the D -dimensional dilution ($1/D = 1/3$) with a geometric mean ($1/2$). Physically, this can be interpreted as: matter's gravitational effect on the horizon is averaged (geometric mean) over its dilution across D spatial dimensions.

3.4 What this decomposition constrains

Any future theoretical derivation of the MOND acceleration from Λ CDM thermal physics must produce a result that matches:

$$a_0 = a_{dS} / (2\pi \times \Omega_\Lambda^{1/3} \times \Omega_m^{1/6}) \quad \dots (7)$$

This provides three specific targets: the derivation must explain why the circumference (2π) rather than the radius is the relevant scale, why Ω_Λ enters at the $1/3$ power, and why Ω_m enters at the $1/6$ power.

4. Relationship to the Thermal Formula

4.1 The empirically motivated substitution

If one replaces a_{dS} with a_0 in Milgrom's thermal formula (Equation 3):

$$T(a) = (\hbar/(2\pi k_B c)) \sqrt{a^2 + a_0^2} \quad \dots (8)$$

the transition scale becomes $a_0 = 1.19 \times 10^{-10} \text{ m/s}^2$ — the correct MOND value. The functional form of ΔT is preserved.

This substitution is not derived. Equation (3) was obtained by Deser & Levin (1997) from a specific calculation in pure de Sitter space. The parameter a_{dS} is not a free parameter — it is the surface gravity of the de Sitter horizon, determined by the metric. In Λ CDM spacetime, the metric differs from de Sitter, and the temperature formula for an accelerated observer has not been computed. The substitution $a_{dS} \rightarrow a_0$ is empirically motivated by the correction factor (Equation 5), not derived from the Λ CDM metric.

We present Equation (8) as a **hypothesis** — a candidate formula whose transition scale matches observations — not as a derived result. Whether the actual Λ CDM temperature formula takes this form is an open problem in quantum field theory in curved spacetime.

We note a subtlety regarding circularity: the correction factor (Equation 5) is the primary result of this paper, and it is purely algebraic. The thermal substitution (Equation 8) is a consistency check — it shows that the corrected formula has the right transition scale — but it is not an independent prediction, since any formula producing the correct a_0 would give the correct transition when substituted. An independent prediction would require the thermal formula to produce a specific interpolation function that matches SPARC data better than alternatives; as discussed in Section 5, this has not been demonstrated.

4.2 The physical picture, if the hypothesis holds

If the substitution is ultimately justified by a Λ CDM calculation, the physical interpretation would be: matter "cools" the effective horizon temperature below the pure de Sitter value. In pure de Sitter space, the horizon's thermal properties are set entirely by the vacuum energy. In Λ CDM, matter's gravitational attraction partially counteracts the accelerating expansion, reducing the effective thermal acceleration from a_{dS} to a_0 — a reduction by a factor of 4.57.

The MOND acceleration temperature would be:

$$T(a_0) = \hbar a_0 / (2\pi k_B c) = 4.81 \times 10^{-31} \text{ K} \quad \dots (9)$$

This is the geometric mean of two horizon temperatures (Nguyen 2026c):

$$T(a_0) = \sqrt{T_{\text{now}} \times T_{\text{eq}}} \quad \dots (10)$$

where $T_{\text{now}} = \hbar c / (k_B \times C_{\text{now}})$ and $T_{\text{eq}} = \hbar c / (k_B \times C_{\text{eq}})$ are the temperatures associated with the current and equality-epoch Hubble circumferences.

4.3 Redshift evolution

If the thermal hypothesis holds, a_0 evolves with redshift because the matter-DE ratio evolves:

$$a_0(z) = cH(z)(\Omega_\Lambda(z)/\Omega_m(z))^{1/6}/(2\pi) \quad \dots (11)$$

This prediction is independent of the thermal interpretation — it follows directly from the empirical formula (Nguyen 2026a). The thermal picture provides a candidate physical mechanism: the effective horizon temperature changes as the cosmic composition evolves. At $z = 1$, a_0 is predicted to be 27% higher; at $z = 2$, 75% higher (Nguyen 2026a).

5. The Interpolation Function and Its Factor-of-2 Problem

5.1 Milgrom's thermal interpolation

If inertia is proportional to $\Delta T/a$ (Milgrom 1999), the implied MOND interpolation function is:

$$\mu(x) = [\sqrt{(1 + x^2)} - 1]/x \quad \text{where } x = a/a_0 \quad \dots (12)$$

Limits:

- $x \gg 1$: $\mu(x) \rightarrow 1 - 1/(2x^2) \rightarrow 1$ (Newtonian) ✓
- $x \ll 1$: $\mu(x) \rightarrow x/2$ (deep-MOND)

5.2 The problem

In standard MOND, the deep-MOND limit requires $\mu(x) \rightarrow x$, giving $a^2 = a_0 g_{\text{bar}}$. The thermal interpolation gives $\mu(x) \rightarrow x/2$, producing:

$$a^2 = 2a_0 g_{\text{bar}} \quad \dots (13)$$

This is a factor of 2 too large. In rotation curve terms, the flat velocity in the deep-MOND limit scales as:

$$v^4 = a_0 GM \text{ (standard MOND)} \quad v^4 = 2a_0 GM \text{ (thermal interpolation)}$$

The thermal formula predicts velocities that are $2^{1/4} = 1.19$ times too high — a 19% systematic excess in every low-surface-brightness galaxy. Given that SPARC fits achieve ~10% scatter, a systematic 19% offset would be detectable and is likely already excluded.

5.3 Status of the problem

This factor-of-2 discrepancy is **not new to this paper**. It is present in Milgrom's original 1999 analysis and has been discussed in the literature on thermal approaches to MOND (see also Volovik 2024, who argues that the local de Sitter temperature is $T = H/\pi$ rather than $H/(2\pi)$, introducing a separate factor-of-2 question). The discrepancy reflects the fact that the Milgrom (1999) framework is "suggestive" rather than rigorous — it identifies the correct functional form but does not produce exact quantitative agreement.

Possible resolutions include:

1. The proportionality between inertia and ΔT involves an additional factor of 2 from degrees of freedom or normalization, absorbing the discrepancy.

2. The correct Λ CDM temperature formula differs from the pure de Sitter formula in a way that eliminates the factor.
3. The thermal framework captures the qualitative physics but not the exact interpolation function, and the true function must be determined empirically.

We do not resolve this issue here. We note it explicitly because it is a known limitation of the thermal approach to MOND that affects any paper — including this one — that connects a_0 to horizon temperatures.

5.4 What survives despite the factor-of-2

The correction factor $a_{\text{dS}}/a_0 = 2\pi \times \Omega_{\Lambda}^{1/3} \times \Omega_{\text{m}}^{1/6}$ (Equation 5) is independent of the interpolation function. It is a pure algebraic relationship between two acceleration scales. Whether the thermal interpolation function reproduces exact MOND is a separate question from whether the ratio between a_{dS} and a_0 has physical meaning. The former is in doubt; the latter is exact.

6. Before and After

6.1 The historical progression of a_0 predictions

The following table summarizes the progression of theoretical estimates of a_0 from cosmological quantities, compared to the McGaugh canonical value $a_0 = 1.20 \times 10^{-10} \text{ m/s}^2$.

Formula	Author(s)	Value ($\times 10^{-10} \text{ m/s}^2$)	Error
cH_0	Milgrom (1983)	6.55	446%
$cH_0\sqrt{\Omega_{\Lambda}}$	Milgrom (1999)	5.42	352%
$\sim cH_0$	Verlinde (2016)	~ 6.55	$\sim 446\%$
$cH_0/(2\pi)$	Famaey & McGaugh (2012)	1.04	13.1%
$cH_0(\Omega_{\Lambda}/\Omega_{\text{m}})^{1/6}/(2\pi)$	Nguyen (2026a)	1.186	1.1%
Observed	McGaugh et al. (2016)	1.20 ± 0.24	—

The Nguyen (2026a) formula brings the prediction within 1.1% of the observed value. For context, the systematic uncertainty on the observed a_0 is $\pm 20\%$ (McGaugh et al. 2016). The remaining 1.1% discrepancy is approximately 17 times smaller than the measurement uncertainty — the prediction and the observation are statistically indistinguishable with current data. We note that the previous best estimate $cH_0/(2\pi)$ is also within the $\pm 20\%$ measurement uncertainty; the single-number comparison in this table does not by itself establish the superiority of the Nguyen formula. The decisive discriminant is the multi-point test against 3,389 SPARC rotation curve measurements ($\Delta\chi^2 = 4,895$ relative to $cH_0/(2\pi)$; see Nguyen 2026a for details).

Each step in the table represents a specific physical insight: Milgrom (1983) identified the cosmological coincidence $a_0 \sim cH_0$. Milgrom (1999) introduced the vacuum energy through $\sqrt{\Omega_\Lambda}$. Famaey & McGaugh (2012) noted the 2π factor. The present work adds the matter correction $(\Omega_\Lambda/\Omega_m)^{1/6}$, which accounts for the fact that our universe contains matter, not just dark energy.

6.2 What changed

The numerical gap between de Sitter thermodynamics and MOND is now exactly quantified: $a_{dS}/a_0 = 2\pi \times \Omega_\Lambda^{1/3} \times \Omega_m^{1/6} = 4.57$. Each component has a candidate physical interpretation: circumference geometry (2π), vacuum energy dimensionality ($\Omega_\Lambda^{1/3}$), and matter dilution ($\Omega_m^{1/6}$). The correction factor constrains any future Λ CDM thermal derivation of MOND: it must produce Equation (7).

6.3 What did not change

The theoretical status of MOND as emergent from horizon thermodynamics remains a hypothesis (Milgrom 1999). The present paper provides the exact correction factor but does not derive it from first principles. The thermal interpolation function's factor-of-2 problem (Section 5) remains open. The substitution $a_{dS} \rightarrow a_0$ in the thermal formula is empirically motivated, not theoretically derived.

7. Discussion

7.1 Implications for theoretical work

The decomposition of the correction factor (Equation 5) provides a roadmap for theorists attempting to derive MOND from Λ CDM thermal physics:

1. **The 2π :** A derivation must explain why the circumference of the cosmological horizon, rather than its radius, sets the MOND scale. In black hole thermodynamics, the Hawking

temperature involves the circumference ($T \propto 1/C(\text{horizon})$). If the MOND-cosmology connection is analogous, the same should hold for the cosmological horizon.

2. **The $\Omega_\Lambda^{1/3}$:** A derivation must explain why the vacuum energy enters at the $1/3 = 1/D$ power. This suggests a dimensional reduction or a per-dimension contribution of the vacuum energy to the thermal state.
3. **The $\Omega_m^{1/6}$:** A derivation must explain why matter enters at the $1/6 = 1/(2D)$ power. The geometric mean ($1/2$) and the dimensional factor ($1/D$) both need physical justification.

7.2 Comparison with other approaches

Verlinde's (2016) emergent gravity also obtains $a_0 \sim cH_0$ without specifying the exact factor. The present result suggests that Verlinde's framework, if it could be made precise in Λ CDM, should produce the factor $(\Omega_\Lambda/\Omega_m)^{1/6}/(2\pi)$.

Klinkhamer's (2011) minimum-temperature approach gives $a_0 \propto cH_0$ through a different route (entropic gravity with a temperature floor). The correction factor identified here could constrain which version of entropic gravity produces the right numerical coefficient.

7.3 The Volovik question

Volovik (2024) argues that the local de Sitter temperature is $T_{\text{local}} = \hbar H/(\pi k_B c)$, which is twice the Gibbons-Hawking temperature $T_{\text{GH}} = \hbar H/(2\pi k_B c)$. This factor-of-2 disagreement in pure de Sitter space is unresolved and is conceptually distinct from the factor-of-2 in the interpolation function (Section 5). If Volovik's local temperature is relevant for the MOND connection, it would change the 2π in Danny's formula to π , producing $a_0 = cH_0(\Omega_\Lambda/\Omega_m)^{1/6}/\pi = 2.37 \times 10^{-10} \text{ m/s}^2$. This is clearly excluded by SPARC data ($a_0 = 1.20 \pm 0.02 \pm 0.24$). The empirical formula therefore selects the Gibbons-Hawking temperature (2π) over the Volovik local temperature (π), if the thermal connection is real.

8. Conclusion

The ratio between the pure de Sitter acceleration scale $a_{\text{dS}} = cH_0\sqrt{\Omega_\Lambda}$ and the empirically measured MOND acceleration $a_0 = cH_0(\Omega_\Lambda/\Omega_m)^{1/6}/(2\pi)$ is exactly:

$$a_{\text{dS}}/a_0 = 2\pi \times \Omega_\Lambda^{1/3} \times \Omega_m^{1/6} = 4.57$$

This closes the numerical discrepancy that Milgrom (1999) identified between the de Sitter thermal framework and the observed MOND transition scale. Each component of the factor has

a candidate physical interpretation rooted in the geometric decomposition of a_0 (Nguyen 2026a,b,c).

The result is algebraic — it follows exactly from the empirical formula. It is not a derivation of a_0 from Λ CDM thermal physics, which remains an open problem. The thermal interpolation function implied by Milgrom's framework has a known factor-of-2 discrepancy with standard MOND that is also not resolved here.

What the result provides is a precise numerical target for future theoretical work: any thermal derivation of MOND in a matter-containing universe must recover Equation (7). The decomposition into circumference, dimensionality, and matter factors identifies the specific features of Λ CDM that such a derivation must incorporate.

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